

Snapshot 3D Gaussian Splatting for Miniature Scenes

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Abstract—We present a snapshot imaging technique for recovering 3D surrounding views of miniature scenes. Due to their intricacy, miniature scenes with objects sized in millimeters are difficult to reconstruct. Yet miniatures are common in life and their 3D digitalization is desirable. We design a catadioptric imaging system with a single camera and multiple pairs of planar mirrors for snapshot 3D reconstruction of miniature scenes from a dollhouse perspective. We first present an in-depth analysis on how to configure catadioptric imaging systems that use a pair of planar mirrors. We derive mirror parameters (e.g., orientation and position) by solving a viewpoint mapping problem, which aims to determine viable mirror configuration that is able to provide reflection image from a desired viewpoint. By applying the design principle, We show a full-surround snapshot catadioptric imaging system built with eight pairs of planar mirrors. Specifically, we place the mirror pairs on nested pyramid surfaces with different angles for capturing surrounding multi-view images in a single shot. This would allow 3D reconstruction of dynamic scenes. Our mirror design is customizable based on the size of the scene for optimized view coverage. We use the 3D Gaussian Splatting (3DGS) representation for scene reconstruction and novel view synthesis. We overcome the challenge posed by our sparse view input by integrating visual hull-derived depth constraint. We perform experiments on rendered synthetic images and real images of a variety of miniature scenes that are captured by our custom-built imaging system. Experimental results demonstrate that our method outperforms state-of-the-art sparse-view and calibration-free 3DGS methods. We also show novel view synthesis results on dynamic miniature scenes (e.g., a live, moving insect).

Index Terms—Catadioptric imaging system, snapshot full-surround imaging, ray geometry analysis, 3D Gaussian splatting, novel view synthesis.

I. INTRODUCTION

MOST existing works on 3D reconstruction or novel view synthesis focus on large-scale or life-size scenes. The tiny world of miniatures (e.g., objects sized in centimeters or even millimeters) is somewhat neglected. 3D reconstruction of miniature scenes is challenging due to limitations on lenses, image resolution and reconstruction accuracy. To take images of miniatures, one would need a macro lens to magnify the tiny objects, such that they could cover substantial amount of pixels in an image. Due to high magnification ratio, macro images usually have shallow depth of field, making it hard to capture all-in-focus image. In addition, many miniatures have

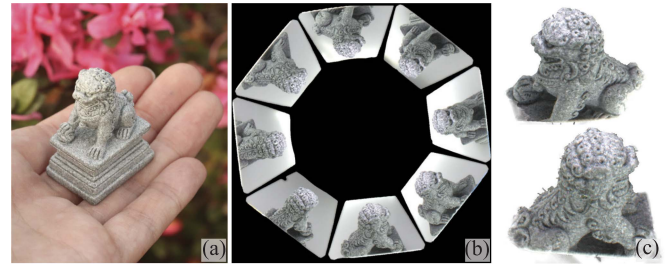


Fig. 1. (a) A miniature scene with tiny object; (b) Example image captured by our catadioptric imaging system; (c) Novel view synthesis results.

little textures (as limited by their size), which poses challenge to photogrammetry-based 3D reconstruction. Yet miniatures scenes are common in life (see example in Fig. 1(a)): from toys and decorations in household to artisanal crafts and antiques in design studios and museums, high-quality 3D reconstruction not only opens up new ways for viewing and interacting with those miniatures, and also benefits their preservation [1].

In this work, we present a snapshot solution for reconstructing miniature scenes and synthesizing their images in full surround 360° views. In order to guide our imaging system design, we first present a principled analysis on how to configure the parameters of catadioptric imaging systems that use a pair of planar mirrors. We formulate the mirror design as a viewpoint mapping problem, which aims to find viable mirror configurations that are able to provide reflection image from a desired viewpoint. We derive the solution space of mirror parameters (e.g., orientation and position) using the mirror transformation matrices.

By following the design principle, we present a novel catadioptric lens with eight pairs of planar mirrors to enable zoom-in all-focus imaging and time-synchronized full surround multi-view acquisition. Specifically, we use the reflection between paired mirrors to fold the light path in order to zoom-in onto small object in short distance. We arrange the eight mirror pairs on the sides of two nested octagonal pyramids to capture surrounding views of the scene in a single shot. An example of our captured image is shown in Fig. 1(b). Since our multi-view images are time-synchronized through mirrors, we are able to perform 3D reconstruction and novel view synthesis on dynamic scenes. We then analyze the ray geometry of our lens and derive closed-form formulas for optimizing the mirror configuration based on the size of the scene, such that the multi-view coverage of the scene is optimal. We map our multi-view sub-images to virtual cameras and pre-calibrate the virtual camera parameters. The calibration only needs to run once when the mirror configuration is set. Since our calibrated camera parameters are highly accurate, they greatly benefit

Received 15 August 2025; revised 31 March 2026; accepted 24 April 2026. Date of publication 29 April 2026; date of current version 6 August 2026. This work was supported by NSF under Grant 2225948 and Grant 2238141. Recommended for acceptance by H. Ling. (Corresponding author: Jinwei Ye.)

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Digital Object Identifier 10.1109/TPAMI.2026.3688999

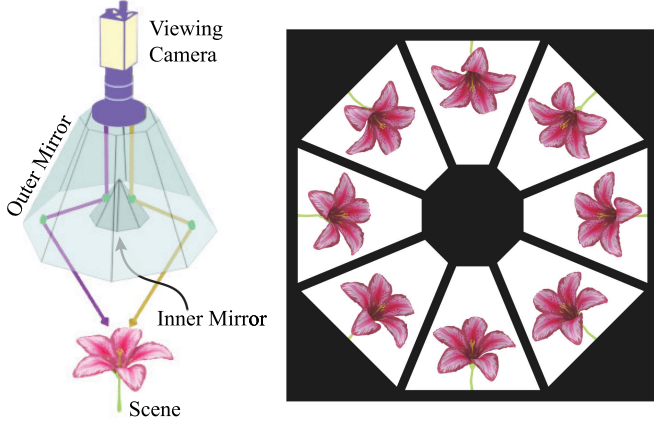


Fig. 2. Left: A schematic illustration of our imaging system; Right: Sample image captured by the system.

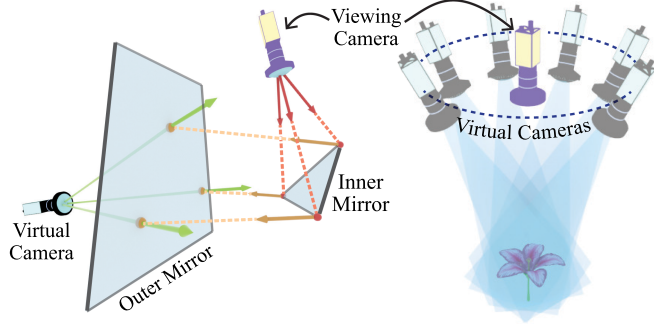


Fig. 3. We map mirror reflections to virtual camera views by unfolding the light paths.

scene reconstruction. In contrast, self-calibration methods (e.g., structure from motion (SfM) [2]) heavily rely on scene features and are not reliable for miniature scenes with little textures.

We use the 3D Gaussian Splatting (3DGS) [3] for scene representation, and synthesize full surround novel views using our multi-view images and pre-calibrated camera parameters. Since our input views are sparse (e.g., 8 views on a 360° circle), we use additional depth constraints derived from visual hull to improve the 3DGS reconstruction. Specifically, we first calculate the object's visual hull using silhouette masks extracted from multi-view images. We then project the visual hull to depth map under each virtual viewpoint. Since visual hull is a convex volume that fully encloses the object, we propose a weighted depth loss that penalizes more on depth values greater than the visual hull depth (which indicates the point is outside of the visual hull). Our depth loss is especially effective for surfaces with little textures, and can be used for generic 3DGS when visual hulls are available.

We evaluate our method on both synthetic and real data. For synthetic experiments, we render images by emulating the configuration of our real imaging system. We qualitatively and quantitatively evaluate our novel view synthesis results by comparing with state-of-the-art sparse-view 3DGS methods. For real experiments, we test on different mirror configurations using 3D-printed housing and acrylic mirrors. We custom-built a compact prototype with precision optics for high-quality imaging.

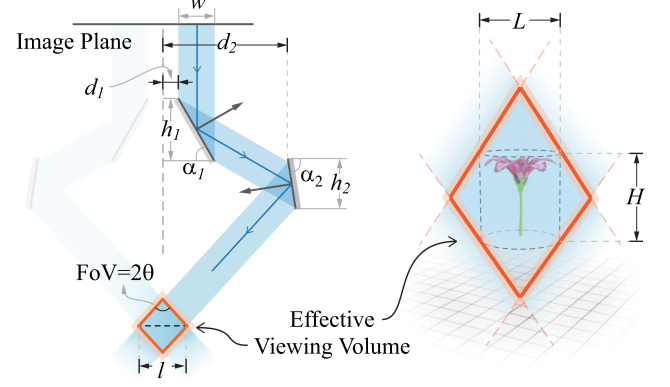


Fig. 4. The reflection light path of our mirror pair and the optimal coverage of a scene by our effective viewing volume.

We capture a variety of miniature scenes in both indoor and outdoor settings. Results show that our method out-performs state-of-the-art sparse-view and calibration-free 3DGS methods. We also show freeze-time novel view synthesis results on several dynamic miniature scenes that are otherwise very challenging to capture and reconstruct (for example, a crawling lady bug).

Here we summarize our main contributions as follows:

- We present a principled analysis on catadioptric system design using a pair of planar mirrors.
- We design a catadioptric imaging system for recovering 3D surrounding views of miniature scenes in a single shot.
- We analyze the ray geometry of our catadioptric lens and derive closed-form formulas for calculating optimized lens parameters given scene information.
- We propose a novel weighted depth loss based on visual hull to improve 3DGS with sparse view input.
- We build a prototype for the proposed imaging system and validate our approach on real miniature scenes.

This work is an extension of our previous publication [4]. In this journal version, we include a principled analysis on paired mirror design, supplement more details about our imaging system configuration and implementation, and add new results on real-captured dynamic scenes.

The remainder of the paper is organized as follows: we first review relevant prior works in Section II. Next, we analyze the design space of catadioptric imaging systems using a pair of planar mirrors in Section III, and then introduce our full-surround catadioptric system in Section IV. We present our 3DGS-based scene reconstruction algorithm in Section V. Section VI shows the experiments we perform with comparison results. Finally, we conclude our work and discuss about our limitations and promising future directions in Section VII.

II. RELATED WORK

Here we briefly review prior works that are most relevant to our imaging system and reconstruction algorithm.

Mirror-based imaging system. Mirrors, either planar or curved, are widely used for building imaging systems with extended field-of-view or snapshot multi-view capacities. This is because mirror reflection provide image from new perspective,

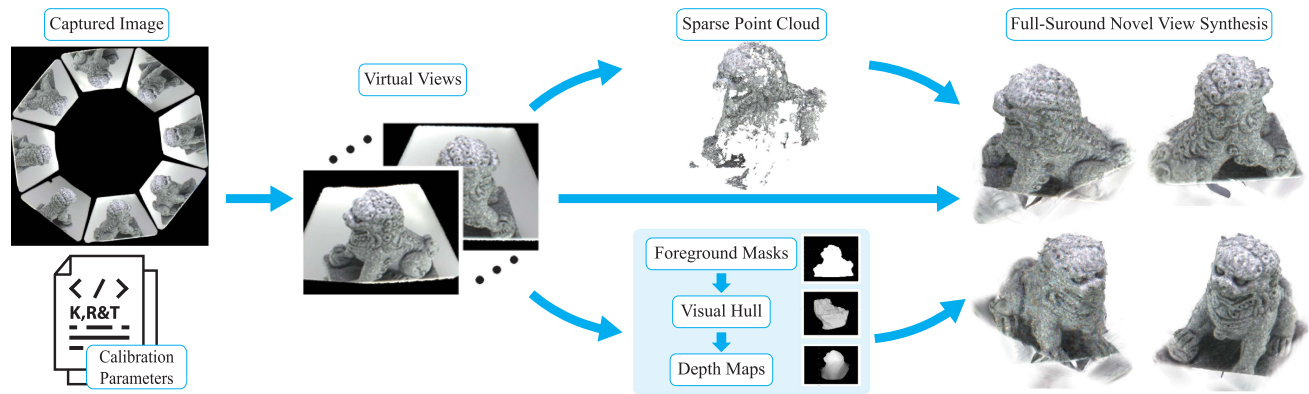


Fig. 5. Our overall full-surround novel view synthesis pipeline.

which is equivalent to having a virtual camera at the new viewpoint. Imaging systems that involve both mirrors and refractive lenses are called catadioptric systems. They have been extensively studied and used for stereo [5], [6], [7], panoramic [8], [9], [10], surround-view [11], [12] and light field [13], [14], [15] imaging. In particular, Nene and Nayar [6] study using various forms of mirrors, including planar, ellipsoidal, hyperboloidal, and paraboloidal ones, to form stereo images and derive their epipolar constraints. Tan et al. [10] investigate the use of mirror pyramid for multi-view panoramic imaging. Fuchs et al. [13] develop a full pipeline for designing, fabricating, and deploying faceted mirror arrays for single-shot light field acquisition with high-resolution sensors. Reshetouski and Ihrke [16] provide a comprehensive survey on the design and applications of various mirror-based imaging systems.

More related to our work, Gluckman and Nayar [17], [18] show all possible configurations for catadioptric stereo and derive the reflection transformation for mirror-based image formation. This work focuses on using mirrors to form rectified stereo image pairs. In contrast, our work investigates the mirror design for more generic viewpoint mapping.

Kaleidoscopic imaging systems use the inter-reflection of mirrors to generate variations in viewpoints and illumination for 3D shape [19], [20] and reflectance [21], [22] reconstruction. Various techniques are explored for 3D reconstruction, including space carving [20], multi-view stereo [23], structured light [11] and neural surface representation [24]. Notably, Ahn et al. [11] design a kaleidoscope-based structured light system for full-surround 3D reconstruction. Since kaleidoscope reflections are resulted by a sequence of mirrors, they use the epipolar geometry between camera and projector to determine the mirror sequence and label virtual cameras. Our work uses eight pairs of mirrors for full-surround imaging and the 3DGS framework for scene reconstruction. In contrast to kaleidoscopes, we avoid inter-reflection in our mirror system. Thus, our multi-view images are much easier to separate and analyze, although the angular resolution is sacrificed.

3DGS with sparse view input. 3D Gaussian Splatting (3DGS) [3] is an efficient radiance field representation that allows for high-quality real-time rendering of novel views, with relatively short training time. The method represents scenes

with 3D Gaussians that retain properties of continuous volumetric radiance fields, such as color and density, by optimizing Gaussian attributes (e.g., position, opacity, covariance, *etc.*) by fitting to a set of reference views of the scene. But the original 3DGS algorithm requires dense view supervision in order to achieve high quality rendering. With few reference views (e.g., less than ten), the algorithm tends to overfit to input views, resulting artifacts in unseen views. The problem may get worse for full-surround view synthesis, since the reference views may have limited overlap. Many recent techniques are proposed to allow robust 3DGS-based novel view synthesis when few views are available. A major trend is to use depth maps to provide additional supervision. GS2Mesh [25] and InstantSplat [26] adopt depth learned from multi-view stereo. ReconFusion [27] and GaussianObject [28] use diffusion model to synthesize additional views in order to provide dense viewpoint and depth supervision. RaDe-GS [29] improves the depth rasterization in 3DGS.

Since the Gaussian attributes are initialized from a sparse point cloud obtained by projecting correspondences in reference views with known camera parameters (i.e., intrinsics and extrinsic poses), it is important to obtain accurate camera parameters for point cloud initialization. Typically, SfM-based methods (e.g., COLMAP [2]) are adopted. There are a class of “pose-free” methods that integrate camera pose estimation with Gaussian optimization. Notably, FreeSplat [30] uses a transformer of sequential self-attention layers to fuse multi-view features and decode them into pixel-wise 3D Gaussian primitives. CF-3DGS [31] sequentially processes input frames and increases the 3D Gaussian set frame-by-frame by leveraging video-frame temporal continuity for pose estimation. InstantSplat [26] uses dense stereo priors and visibility relationships between consecutive frames to initialize the scene geometry in a progressive manner. MV-DUST3R+ [32] employs cross-reference-view blocks to process multiple views together and uses the output for Gaussian optimization. In this work, we pre-calibrate the parameters of our virtual cameras through a specially designed target, and use highly-accurate camera parameters for point cloud initialization and Gaussian optimization. To address the sparse view challenge, we propose a new weighted depth loss that uses visual-hull-generated depth maps to improve the quality of synthesized novel views.

III. CATADIOPTRIC SYSTEM DESIGN USING PAIRED MIRRORS

In this section, we present a principled analysis on catadioptric system design using a pair of planar mirrors. Since catadioptric systems provide multi-view images in a single shot, we formulate the mirror parameter design as a viewpoint mapping problem: how to find viable mirror configurations that can provide multi-view images from desired viewpoints? As the viewpoint transformation allowed by one mirror is rather limited, we consider using two mirrors, putting together in a pair (i.e., form image through two consecutive mirror reflections), in order to enable viewpoint transformation with higher degree of freedom (DoF).

Specifically, we aim at solving mirror parameters by satisfying a set of given viewpoint constraints, defined in terms of rotation and translation among viewpoints. Viable solutions of the viewpoint mapping problem then provide the design space for practical mirror configuration.

Mirror transformation background. We first give a brief review of mirror transformation and examine the space of viewpoint transformation allowable by a single mirror and a pair of mirrors.

When imaging with a single mirror, one can obtain a reflection image that is taken by a virtual camera positioned at the other side of the mirror, symmetric to the real viewing camera. As derived in [17], the mirror reflection transformation matrix can be written as:

$$D = \begin{pmatrix} I - 2\mathbf{n}\mathbf{n}^\top & 2l\mathbf{n} \\ \mathbf{0} & 1 \end{pmatrix}, \quad (1)$$

where I is a 3×3 identity matrix; $\mathbf{n} \in \mathbb{R}^{1 \times 3}$ is a column vector that represents the mirror's normal; l is the distance between the real camera's center of projection (CoP) and the mirror (measured as the length of a perpendicular line dropped from the CoP to the mirror plane).

The reflection matrix D transforms the viewpoint¹ of the real camera to a virtual one. It can be viewed as combination of a rigid transformation and a flip of the camera coordinate system. Let the top-left 3×3 sub-matrix of D be $R = I - 2\mathbf{n}\mathbf{n}^\top$, and the last 1×3 column vector be $\Delta\mathbf{t} = 2l\mathbf{n}$. R combines rotation and coordinate flipping (e.g., sign flipping), and $\Delta\mathbf{t}$ indicates the translation of viewpoint. Therefore, we can see that when the rotation between two viewpoints are specified (i.e., fixed R), then the CoP can only be translated along the normal direction, since $\mathbf{t}$ and $\mathbf{n}$ have the same direction.

When imaging with two mirrors in pair, the virtual camera is transformed by two consecutive reflection transformations. Let's use M_1 & M_2 to denote the two mirrors. WLOG, we assume incoming light first reflects by M_1 , and then by M_2 , before entering the camera. By applying (1) on M_1 & M_2 , the paired mirror transformation can be calculated as:

$$D_{\text{pair}} = D_2 D_1 = \begin{pmatrix} R_2 R_1 & R_2(2l_1\mathbf{n}_1) + 2l_2\mathbf{n}_2 \\ \mathbf{0} & 1 \end{pmatrix}, \quad (2)$$

¹We define viewpoint in terms of the CoP's location and the sensor's orientation.

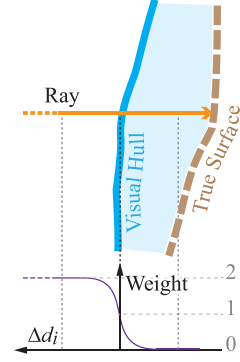


Fig. 6. S-shape weight.

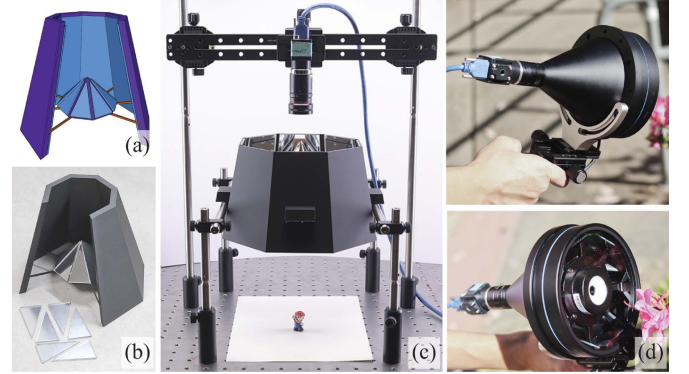


Fig. 7. Prototypes of our mirror lens. (a) 3D model of our lens housing; (b) 3D printed housing with acrylic mirror pairs adhered to the inside; (c) Our imaging setup; (d) Our high-precision portable prototype.

where $\mathbf{n}_1$ and $\mathbf{n}_2$ are the normals of M_1 & M_2 respectively; l_1 and l_2 are the distances between the real camera's CoP and two mirror planes; $R_1 = I - 2\mathbf{n}_1\mathbf{n}_1^\top$ and $R_2 = I - 2\mathbf{n}_2\mathbf{n}_2^\top$.

Unlike the single mirror transformation that flips the camera coordinate, the paired mirror transformation is a pure rigid transformation in form of rotation ($R_2 R_1$) and translation ($R_2(2l_1\mathbf{n}_1) + 2l_2\mathbf{n}_2$). This is because the camera coordinate is flipped twice by the mirrors. As a result, the virtual camera coordinate has the same handedness as the real camera.

By applying the Rodrigues' formula, we can derive the axis of rotation as $\mathbf{k} = \mathbf{n}_1 \times \mathbf{n}_2$ [33], and the angle of rotation as $\Phi = 2\varphi$, where $\varphi = \arccos(\mathbf{n}_1^\top \mathbf{n}_2)$ is the angle between the two mirrors' normals. Please see the appendix, available online, for detailed derivation.

Since the translation vector is a linear combination of $\mathbf{n}_1$ and $\mathbf{n}_2$, it must lie on the plane spanned by $\mathbf{n}_1$ and $\mathbf{n}_2$. Given a fixed rotation (i.e., $R_2 R_1$), the normal spanned plane is determined since it should be perpendicular to the rotation axis $\mathbf{k}$. Therefore, a pair of mirrors would allow translation on a 2D plane (i.e., 2 DoF). Since the rotation can be arbitrary (i.e., 3 DoF), a pair of mirrors allows 5 DoF viewpoint transformation. In order to achieve the full 6 DoF viewpoint transformation, three mirrors should be used. However, three mirrors may result in occlusions that make all possible configurations infeasible, and their transformation will cause flipping of the camera coordinate that is undesired by many downstream applications. Therefore here we focus on analyzing the design using two mirrors.

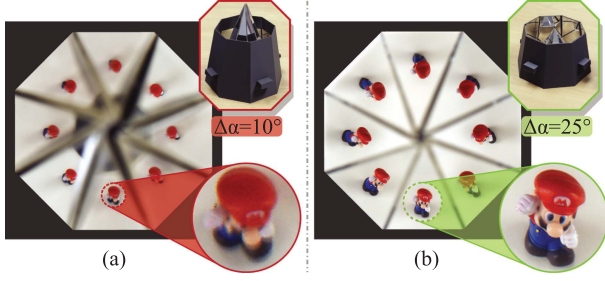


Fig. 8. Comparison of two lens prototypes with different mirror angles, where the angular difference of (a) is 10 degree and (b) is 25 degree. Here we show the lens prototypes and their captured images with zoom-in views.

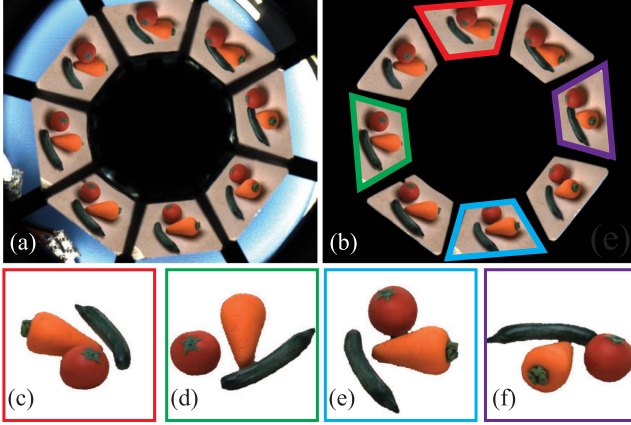


Fig. 9. (a) Our captured raw image; (b) Image filtered by the multi-view mask; (c)-(f) Re-projected multi-view images (border colors indicate correspondences in the all-in-one image).

Mirror design as viewpoint mapping problem. Given the mirror transformations, now we formulate the paired mirror design problem as a viewpoint mapping problem: how to design mirror parameters (i.e., orientation, location, and size of mirrors) in order to obtain a reflection image from desired viewpoint? This would allow us to use paired mirrors to achieve controlled viewpoint sampling.

The viewpoint mapping problem is defined as follows. WLOG, we consider the problem under the real viewing camera's coordinate. Assume that we want to obtain image from a virtual viewpoint with extrinsic parameters $(R_v, \mathbf{t}_v)$, our goal is to determine viable configuration of a pair of mirrors that can achieve this viewpoint transformation. The mirror configuration can be parameterized by their normals and distances to the real camera: $(\mathbf{n}_1, l_1)$ & $(\mathbf{n}_2, l_2)$.

We first convert the virtual camera's rotation matrix R_v to the angle and axis of rotation. By applying the Rodrigues' formula, we have:

$$\begin{aligned} \text{Rotation angle: } \Phi &= \arccos\left(\frac{\text{tr}(R_v) - 1}{2}\right), \\ \text{Rotation axis: } \mathbf{k} &= \frac{1}{2 \sin \Phi} \begin{pmatrix} R_v[3, 2] - R_v[2, 3] \\ R_v[1, 3] - R_v[3, 1] \\ R_v[2, 1] - R_v[1, 2] \end{pmatrix}, \end{aligned} \quad (3)$$

where $\text{tr}(\cdot)$ takes the trace of a matrix; $R_v[i, j]$ denotes an element in R_v at the i -th row and j -th column.

By comparing with the paired mirror transformation, we can determine the relative orientation between the two mirrors with Φ and $\mathbf{k}$. The angle spanned by the mirrors' normals is half of the rotation angle: $\varphi = \Phi/2$, and the two normals lie on a plane π that is perpendicular to the rotation axis $\mathbf{k}$: $\mathbf{k}^\top \mathbf{n}_1 = 0$ and $\mathbf{k}^\top \mathbf{n}_2 = 0$. Note that viable normals are unique, as they can rotate as a set on π with angle φ in between. If we fix $\mathbf{n}_1, \mathbf{n}_2$ can be calculated using the Rodrigues' formula as:

$$\mathbf{n}_2 = \cos \varphi \mathbf{n}_1 + \sin \varphi (\mathbf{k} \times \mathbf{n}_1). \quad (4)$$

By applying (2) and substituting R_2 , we can correlate the desired translation $\mathbf{t}_v$ with mirror parameters:

$$\begin{aligned} \mathbf{t}_v &= R_2(2l_1 \mathbf{n}_1) + 2l_2 \mathbf{n}_2, \\ &= (I - 2 \mathbf{n}_2 \mathbf{n}_2^\top)(2l_1 \mathbf{n}_1) + 2l_2 \mathbf{n}_2, \\ &= 2l_1 \mathbf{n}_1 + (-4 \cos \varphi l_1 + 2l_2) \mathbf{n}_2. \end{aligned} \quad (5)$$

Note that $\mathbf{n}_1^\top \mathbf{n}_2 = \mathbf{n}_2^\top \mathbf{n}_1 = \cos \varphi$. By projecting $\mathbf{t}_v$ onto $\mathbf{n}_1$ and $\mathbf{n}_2$, we can obtain two linear equations about l_1 and l_2 :

$$\begin{cases} \mathbf{n}_1^\top \mathbf{t}_v = (2 - 4 \cos^2 \varphi) l_1 + 2 \cos \varphi l_2, \\ \mathbf{n}_2^\top \mathbf{t}_v = -2 \cos \varphi l_1 + 2 l_2. \end{cases} \quad (6)$$

Then we can solve l_1 and l_2 as:

$$\begin{cases} l_1 = \frac{(\mathbf{n}_1^\top - \cos \varphi \mathbf{n}_2^\top)}{2 \sin^2 \varphi} \mathbf{t}_v, \\ l_2 = \frac{(\cos \varphi \mathbf{n}_1^\top - \cos 2\varphi \mathbf{n}_2^\top)}{2 \sin^2 \varphi} \mathbf{t}_v. \end{cases} \quad (7)$$

The above derivation indicates that once $\mathbf{n}_1$ is fixed, all other mirror parameters (i.e., $\mathbf{n}_2, l_1$, and l_2) can be uniquely determined. The minimum sizes of the two mirrors can be calculated using the real viewing camera's field of view, given the mirror parameters. When varying $\mathbf{n}_1, \mathbf{n}_2$ will be rotated accordingly (as their spanned angle is a fixed value φ), and the mirrors' locations and sizes will be changed and can be calculated using (7).

Equations (4) and (7) give the design space of a pair of mirrors when a desired virtual viewpoint is specified (i.e., given R_v and $\mathbf{t}_v$). Although the solution of mirror parameters is not unique, the full set parameters are formulated as functions of $\mathbf{n}_1$ and the desired viewpoint's extrinsics. $\mathbf{n}_1$ can be further decided based on practical system design requirements, such as smaller mirror sizes or more compact spacing between mirrors.

IV. FULL-SURROUND PAIRED MIRROR SYSTEM

In this section, we introduce our imaging system for snapshot full-surround miniature scene reconstruction. We first describe the design of our planar catadioptric lens, and then analyze the ray geometry of the lens and derive optimized lens parameters given scene information.

A. Catadioptric Lens Design

Our imaging system consists of a camera and a planar catadioptric lens that allows for synchronized full-surround multi-view image acquisition. The conceptual design is illustrated in

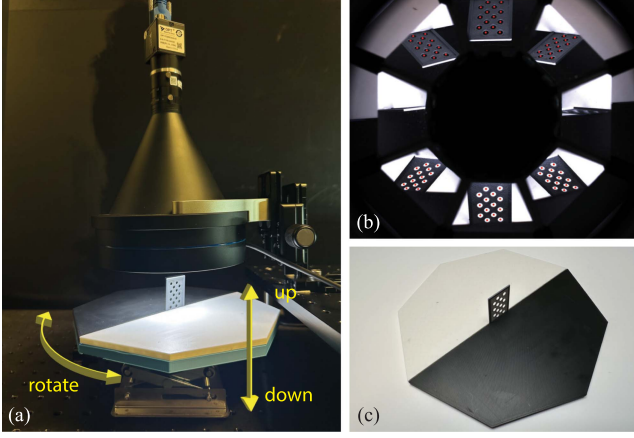


Fig. 10. (a) Our calibration setup; (b) Captured image of calibration board with circle center detected; (c) Our 3D-printed specialized calibration target.

TABLE I
QUANTITATIVE COMPARISON RESULTS ON SYNTHETIC DATA

Method	SSIM $\uparrow$	PSNR $\uparrow$	LPIS $\downarrow$
Hierarchical 3DGS	0.9750	26.8259	0.0298
FSGS	0.7844	18.9281	0.1100
DNGaussian	0.9128	21.3979	0.1296
SparseGS	0.9756	31.8415	0.0367
Ours	0.9783	32.4792	0.0265

TABLE II
ABLATION STUDY ON DEPTH LOSS. HERE WE SHOW THE PSNR OF NOVEL VIEW SYNTHESIS RESULTS OF DIFFERENT VARIANTS.

Scene	SparseGS		Our methods		
	w. mono	w. VH	w/o depth	w. mono	w. VH
Skull	25.4583	27.6893	29.2052	29.3698	29.3856
Squirrel	32.4648	33.9522	34.4545	34.5228	34.5250
Flower	31.3207	31.3085	33.3779	33.6519	33.5773

Fig. 2. By following our design principle, we arrange eight pairs of planar mirrors on a circle to achieve full-surround imaging. In each pair, the mirrors are facing each other with different tilting angles and shifted locations. We therefore arrange the mirrors on the sides of two nested octagonal pyramids. We can adjust the tilting angle and mirror spacing to achieve desired virtual viewpoint. Through two times of reflections, the mirror pair guides light from a scene that is underneath to the viewing camera on top. In this way, the light paths are folded such that we are able to image small objects in short distance, as if they are imaged by a zoom lens. Similar idea on light path folding is used in Folded Optics (or Origami Optics) [34], [35] for compact wide-angle lens design.

Then we arrange eight such mirror pairs along a circle, on surfaces of two nested octagonal pyramids, in order to capture full-surround multi-view images. A sample image captured by our imaging system is shown in Fig. 2. Each sub-image is formed by a pair of mirrors. With eight mirror pairs, we obtain eight sub-images that surround the scene in 360° and are time-synchronized by optics. Note that we purposely arrange the mirrors such that light only bounces once on a piece of mirror. Therefore, there is no inter-reflection in our sub-images. Similar design with mirror pairs is studied for stereo image

acquisition [36]. But two mirror pairs on opposite sides have non-overlapping views and is not practical for stereo vision, whereas our circular arrangement guarantees overlapping in neighboring views and provides full-surround coverage of the scene.

Mapping to virtual cameras. By unfolding the reflection light paths, we are able to map our sub-images to virtual camera views (as if there are no mirrors). This unfolding process and mapped virtual cameras are illustrated in Fig. 3. The virtual cameras are evenly distributed on a circle, looking inward to the scene in the center. All cameras share the same intrinsic parameters, since mirror reflection linearly maps an image. The camera poses are determined by the mirror parameters (e.g., tilting angles and positions). By varying the mirror configuration, we are able to adjust the viewing directions and baseline of surrounding views, so as to optimize their coverage of the scene. In Section IV-B, we analyze the multiview coverage in relationship to the mirror configuration in details.

Comparison with kaleidoscopic imaging. Kaleidoscopic imaging systems use the inter-reflection of mirror chambers for snapshot surrounding view acquisition [11], [21]. One key distinction between our design and kaleidoscopes is that we only allow one-bounce reflection on mirrors and so we do not have inter-reflection in images. With simpler ray geometry, our system is much easier to calibrate and we mitigate the challenging “labeling” problem on determining the mirror sequence involved in inter-reflections. However, the angular resolution is traded off in our design: we have fewer viewing directions comparing to kaleidoscope since the inter-reflected views are absent. Nevertheless, we show in Section V and through experiments that our sparse surrounding views are sufficient for 3DGS-based reconstruction and full surround novel view synthesis.

B. Ray Geometry Analysis

Next we analyze the light transport inside of our lens and derive the relationship between our viewing volume and mirror parameters. We then use the formulas to optimize the mirror configuration based on the scene dimensions, such that the multi-view coverage of the scene is optimal.

Effective viewing volume. The effective viewing volume of our imaging system is the intersection of viewing frustums of all virtual cameras. Here we derive the relationship between the dimension of effective viewing volume and the mirror parameters.

Since our sub-images exhibit weak perspective with long effective focal length, we assume orthographic camera model for simplicity. The effective viewing volume is carved out by eight beams of parallel light as a polyhedron that can be approximated as a symmetric double pyramid. We characterize this volume by its side angle at apex (θ) and the widest length at the base (l). WLOG, we illustrate the formation of effective viewing volume in a 2D cross-section (see Fig. 4), since our mirror design is symmetric. We parameterize the mirrors in a pair using their tilting angles ($\alpha_{1,2}$), vertically projected heights ($h_{1,2}$), and distances ($d_{1,2}$) between their upper edges and the central ray (i.e., vertical ray from the imager center). We denote the

inner mirror as M_1 and the outer one as M_2 . Subscript of the parameters indicate their correspondence to the mirrors.

Since light path is reversible, we trace rays from the camera to the scene. Given a vertical ray from camera, it is first bounced on M_1 and then on M_2 before reaching the scene. The ray's incident angle to M_1 (i.e., the angle between the incident ray and the mirror's normal) is: $\omega_1 = \alpha_1$. It's incident angle to M_2 can then be calculated as: $\omega_2 = 2\alpha_1 - \alpha_2$. Since rays reflected from M_2 intersect with the symmetric beam from the opposite side to form the effective viewing volume, the pyramid's side angle at base can be calculated as: $\beta = 90^\circ - \alpha_2 + \omega_2 = 90^\circ - 2(\alpha_2 - \alpha_1)$. The side angle at apex is thus:

$$\theta = 90^\circ - \beta = 2(\alpha_2 - \alpha_1). \quad (8)$$

Let $\Delta\alpha = \alpha_2 - \alpha_1$. We define $2\theta = 4\Delta\alpha$ as the field-of-view (FoV) of our catadioptric lens. Detailed derivation of these angles can be found in the appendix, available online.

The width of parallel light beam that can enter the camera is determined by M_1 's size and angle: $w = h_1 / \tan \alpha_1$. Since w remains constant after reflection, we can calculate the base length l of the viewing volume as:

$$l = \frac{w}{\cos \theta} = \frac{h_1}{\tan \alpha_1 \cdot \cos 2\Delta\alpha}. \quad (9)$$

Discussions on mirror design. From (8) and (9), we can see that the dimension of effective viewing volume is determined by the angle and size of M_1 (h_1 and α_1) and the angle difference between M_1 and M_2 ($\Delta\alpha$). The greater the $\Delta\alpha$, the larger the FoV. In order to form reflection images from underneath without inter-reflection, the mirror parameters need to satisfy the following three conditions:

- i) $45^\circ < \alpha_1 < \alpha_2 < 90^\circ$, such that light from scene underneath can be reflected to the imager on top;
- ii) $h_2 \geq \frac{\sin \alpha_2}{\tan \alpha_1 \cdot \cos(\alpha_2 - 2\alpha_1)} \cdot h_1$, such that M_2 can cover the entire light beam reflected from M_1 ;
- iii) $d_2 \geq \frac{\tan \alpha_1 + \cot 2\Delta\alpha}{\tan \alpha_1 \cdot (\cot 2\Delta\alpha - \cot 2\alpha_1)} \cdot h_1 + d_1$, such that no inter-reflection occurs between M_1 and M_2 .

Please see the appendix, available online, for derivation of these conditions. Note that although the size and location of M_2 (h_2 and d_2) would not affect the dimension of effective viewing volume dimension, changing these parameters would result in the viewing volume to be shifted vertically. It is preferable for d_2 to take smaller values, since the resulting viewing volume would be closer to the camera.

Optimized mirror configuration. We then show how to find optimal configuration for the mirror pair, given the size of the scene. Since mirror parameters determine the dimension of effective viewing volume. We first find a viewing volume that provides the optimal scene coverage, and then use (8) and (9), along with the three conditions to calculate mirror parameters. We use the following two criteria to find the optimal viewing volume: 1) in order to have a complete reconstruction of the scene, the viewing volume should be large enough to fully enclose the scene; 2) larger FoV is preferred in order to have more coverage on the lateral sides. Since FoV is intersected by

the viewing directions of virtual cameras, the larger the FoV, the more oblique the viewing angles are, and thus the more coverage on the side.

Since the amount of light that can be received by the imager is bounded by its size, we first allow the width of the parallel beam w to take its maximum value, which is half of the sensor size,² and we denote it as $w_{\max}$. Since $w = h_1 / \tan \alpha_1$, we can adjust the angle and length of M_1 to achieve $w_{\max}$. Given $w_{\max}$, the dimension of effective viewing volume is only related to $\Delta\alpha$. The greater the $\Delta\alpha$, the larger the FoV and base length l , but the smaller the vertical height $h = w_{\max} / \sin 2\Delta\alpha$. We then set out to find the viewing volume with the largest FoV that can fully enclose a scene with known size. We approximate a scene using its bounding box with size $W \times L \times H$ (where H is the vertical height and we assume $L \geq W$). The FoV takes its largest value when the scene's bounding box is inscribed in the viewing volume (see Fig. 4). This is because when FoV further increases beyond this value, the vertical height h decreases, which would result in the scene being partially outside of the viewing volume.

Next we derive the FoV value under this circumstance. With similar triangles, we have $(h - H)/h = L/l$. By substituting $h = w_{\max} / \sin 2\Delta\alpha$ and $l = w_{\max} / \cos 2\Delta\alpha$, we have $H \cdot \sin 2\Delta\alpha + L \cdot \cos 2\Delta\alpha = w_{\max}$. By applying the sine angle addition identity, we have:

$$\Delta\alpha = \frac{1}{2} \left(\arcsin \left(\frac{w_{\max}}{\sqrt{L^2 + H^2}} \right) - \arctan \left(\frac{L}{H} \right) \right). \quad (10)$$

The largest FoV is $4\Delta\alpha$, which can be calculated given the scene size. Equation (10) indicates the angle difference between M_1 and M_2 , which is the optimal mirror configuration to achieve the largest FoV.

V. SCENE RECONSTRUCTION

In this section, we show how to use our capture image for scene reconstruction and novel view synthesis. Specifically, we adopt the 3D Gaussian Splatting (3DGS) framework. We use depth supervision constrained by visual hull to improve the quality of full-surround novel view synthesis with sparse reference views. The overall algorithmic pipeline of our method is shown in Fig. 5.

A. Pre-Processing Steps

Since our multi-view images are combined into a single image, we first segment our captured image into individual multi-view images. We obtain the segmentation mask by taking one image of a solid color background. By applying the mask, we obtain eight images as if captured by surrounding virtual cameras. Since the orientation of virtual cameras are flipped on opposite sides, we re-project the multi-view images onto new viewing planes, such that the transition of view poses is smooth on a circle. We also create foreground masks to segment out the object of interest. This can be done using off-the-shelf

²We assume orthographic model. Under perspective model, this value also depends on the lens.

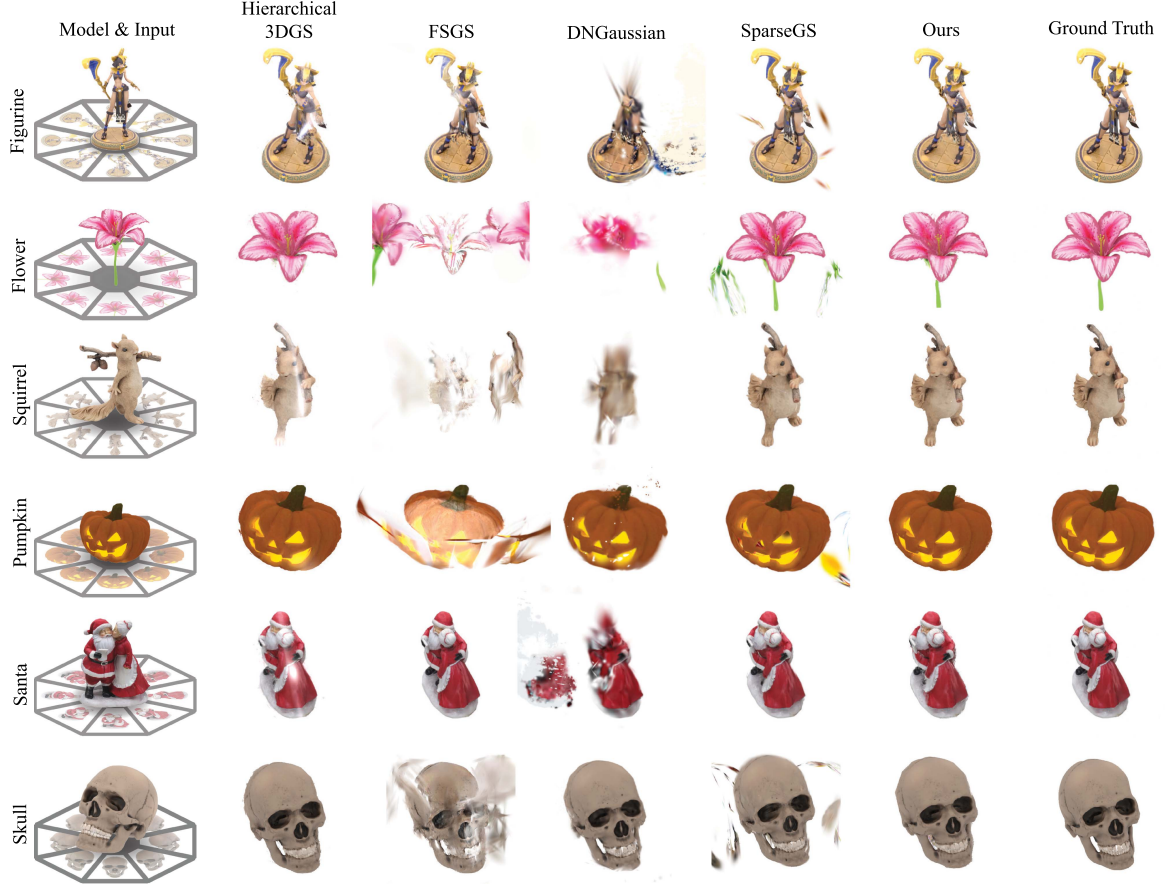


Fig. 11. Visual comparison results on synthetic data. See appendix, available online, for more results and videos of full-surround rendering.

segmentation tools, such as Segment Anything [37]. Please see examples of our input and processed images in Section VI-B.

We also pre-calibrate our imaging system using a special 3D-printed circle grid patterns target to obtain the intrinsic and extrinsic parameters of virtual cameras. Please see Section VI-B for more details of our special calibration board. The extrinsic parameters are re-calculated after view re-projection. The calibration process only needs to run once when our catadioptric lens is fixed on the camera. Pre-calibrating these parameters greatly benefit miniature scene reconstruction, since these tiny scenes often lack textures, and this would result in self-calibration methods (e.g., COLMAP [2]) to fail. With the camera parameters, we estimate an initial point cloud using structure from motion. We then use the eight multi-view images, along with the camera parameters and the initial point cloud as input for 3DGS-based scene reconstruction.

B. 3DGS Representation

3DGS, first introduced by Kerbl et al. [3], represents a scene using a set of 3D Gaussian elements. It is a more compact and efficient 3D representation, comparing to NeRF [38] and classical triangle meshes. It has been demonstrated great success in many rendering applications [39], [40], [41].

A 3D Gaussian is defined by its center position $\mu \in \mathbb{R}^3$ and a covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$:

$$G(\mathbf{x}) = e^{-\frac{1}{2}(\mathbf{x}-\mu)^\top \Sigma^{-1}(\mathbf{x}-\mu)}, \quad (11)$$

where $\mathbf{x}$ is a point on the Gaussian, and Σ can be further decomposed into rotation and scaling matrices (R and S respectively): $\Sigma = RSS^\top R^\top$.

To render an image, the 3D Gaussians are projected by: $\Sigma' = JW\Sigma W^\top J^\top$, where W is the viewing transformation matrix and J is the Jacobian of the affine approximation of the projective transformation. A pixel's color $\mathbf{C}$ is calculated by blending the color of ordered Gaussians that overlap the pixel, similar to the NeRF-style rendering:

$$\mathbf{C} = \sum_{i=1}^n c_i \alpha_i T_i, \quad (12)$$

where c_i is the color of the Gaussian, α_i is the opacity of the 2D projected Gaussian and $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ is the transmittance along the ray.

C. Optimization

To optimize the Gaussian parameters, we compare the rendered output against our input reference views using the following loss function:

$$\mathcal{L} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_{\text{D-SSIM}} + \lambda_3 \mathcal{L}_{\text{depth}}, \quad (13)$$

where $\lambda_{1,2,3}$ are weighting factors for balancing the terms. We use $\lambda_1 = 0.8$, $\lambda_2 = 0.2$ and $\lambda_3 = 0.5$ in our experiments. $\mathcal{L}_1$ and $\mathcal{L}_{\text{D-SSIM}}$ are standard losses that evaluate the color similarity between the rendered image and the reference image. Since our

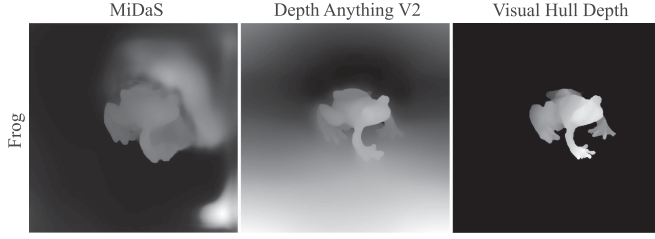


Fig. 12. Comparison of depth map obtained by different methods.

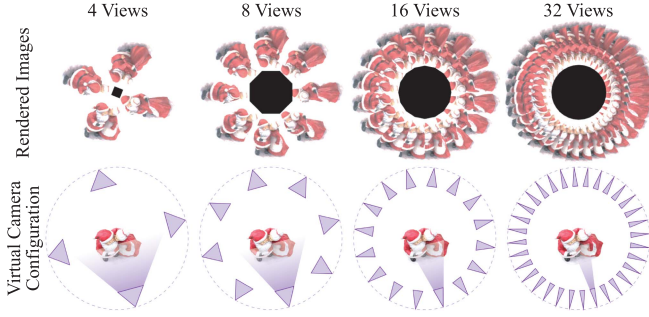


Fig. 13. Trade-off between the spatial and angular sampling of virtual views w.r.t. the total number of views. Based on the size of our scenes, we choose to use 8 views as a balanced configuration.

input views are sparse with limited overlapping, using the color losses alone results in severe artifacts when synthesizing unseen views. We therefore add the third loss $\mathcal{L}_{\text{depth}}$ (which will be described in the following paragraph), a depth loss constrained by visual hull-generated depth maps, to improve the rendering quality of novel views.

Visual hull-constrained depth loss. By using the camera parameters and foreground object masks (which indicate their silhouettes), we can carve out a visual hull [42] for objects in the scene. We then project the visual hull under each reference view and render depth map $\mathbf{D}_{\text{VH}}$ for depth regularization.

We use the rasterizer provided by [43] to render depth map $\mathbf{D}_{\text{render}}$ from 3D Gaussians. Specifically, the depth is rendered by replacing the color c_i in (12) with the depth d_i of the Gaussian's center: $\mathbf{D}_{\text{render}} = \sum_{i=1}^n d_i \alpha_i T_i$. Since $\mathbf{D}_{\text{render}}$ and $\mathbf{D}_{\text{VH}}$ have consistent scale, we calculate the depth loss $\mathcal{L}_{\text{depth}}$ as a weighted L1 norm:

$$\mathcal{L}_{\text{depth}} = \frac{2}{1 + e^{\Delta d_i}} |\mathbf{D}_{\text{render}} - \mathbf{D}_{\text{VH}}|, \quad (14)$$

where $\Delta d_i = \mathbf{D}_{\text{render}}(\mathbf{p}_i) - \mathbf{D}_{\text{VH}}(\mathbf{p}_i)$, with $\mathbf{p}_i$ refers to a pixel in image.

$\mathcal{L}_{\text{depth}}$ is weighted by a s-shaped logistic function with values greater than 1 when $\Delta d_i > 0$ (see Fig. 6). We introduce this weight based on the observation that visual hull is a convex enclosure of the actual geometry. Therefore, we use larger weight for points that are outside of the visual hull (i.e., $\Delta d_i > 0$). For points inside of the visual hull, we gradually decrease the weight to zero, since its depth may still be correct even though $\Delta d_i \neq 0$, if the surface is concave. This weight allows us to model the visual hull-constrained depth regularization more precisely.

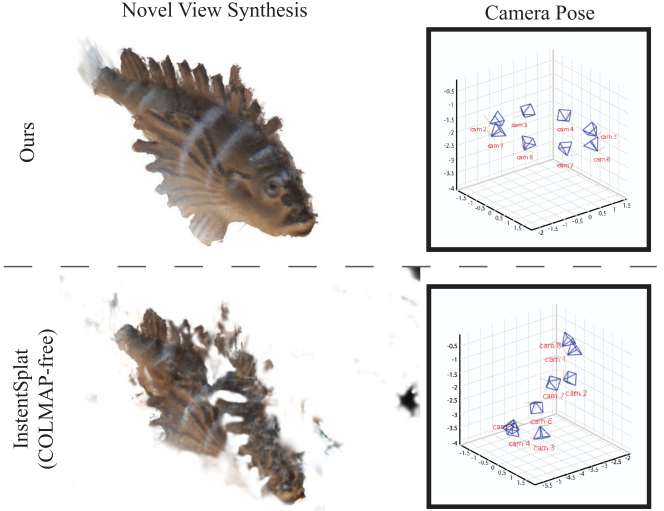


Fig. 14. Comparison with a sparse view COLMAP-free method [26].

VI. EXPERIMENTS

We evaluate our method on both synthetic and real-captured data. Our codes and input data are available on our project website.³

A. Experimental Setup

Synthetic data generation. We simulate synthetic images as captured by our proposed catadioptric imaging system using Autodesk 3ds Max. We build 3D models of mirror pairs and render images using a viewing on top of the mirror pyramid. The mirror parameters are similar to our custom-built real prototype (as shown in Fig. 7(d)). We render 15 scenes with various levels of complexity in terms of texture and geometry. We render ground truth masks for segmenting the multi-view images and foreground objects. In order to calibrate the virtual cameras, we render checkerboard images with different poses. We use ground truth intrinsic parameters and only calibrate the extrinsic parameters. We also render ground truth novel view images for quantitative evaluation. Since our 8 virtual cameras are arranged on a circle (which are our input reference views), we render 24 views along the same circle and use them as ground truth for evaluating novel view synthesis results.

Real experiment setup. We build prototypes for our proposed mirror lens in order to perform real experiments. In order to validate our derivation on mirror angle optimization, we custom-build lens with different mirror angles using 3D-printed housing and acrylic mirrors. The lens 3D model and prototype components are shown in Fig. 7(a) and (b). We mount a 5 megapixel FLIR camera with 16 mm lens on top of the lens to capture images of a scene that is place below the mirror lens. Since our lens block light from above the scene, we mount LED light underneath the inner pyramid in order for the scene to be well lit.

³<https://miniature-3dgs.github.io>

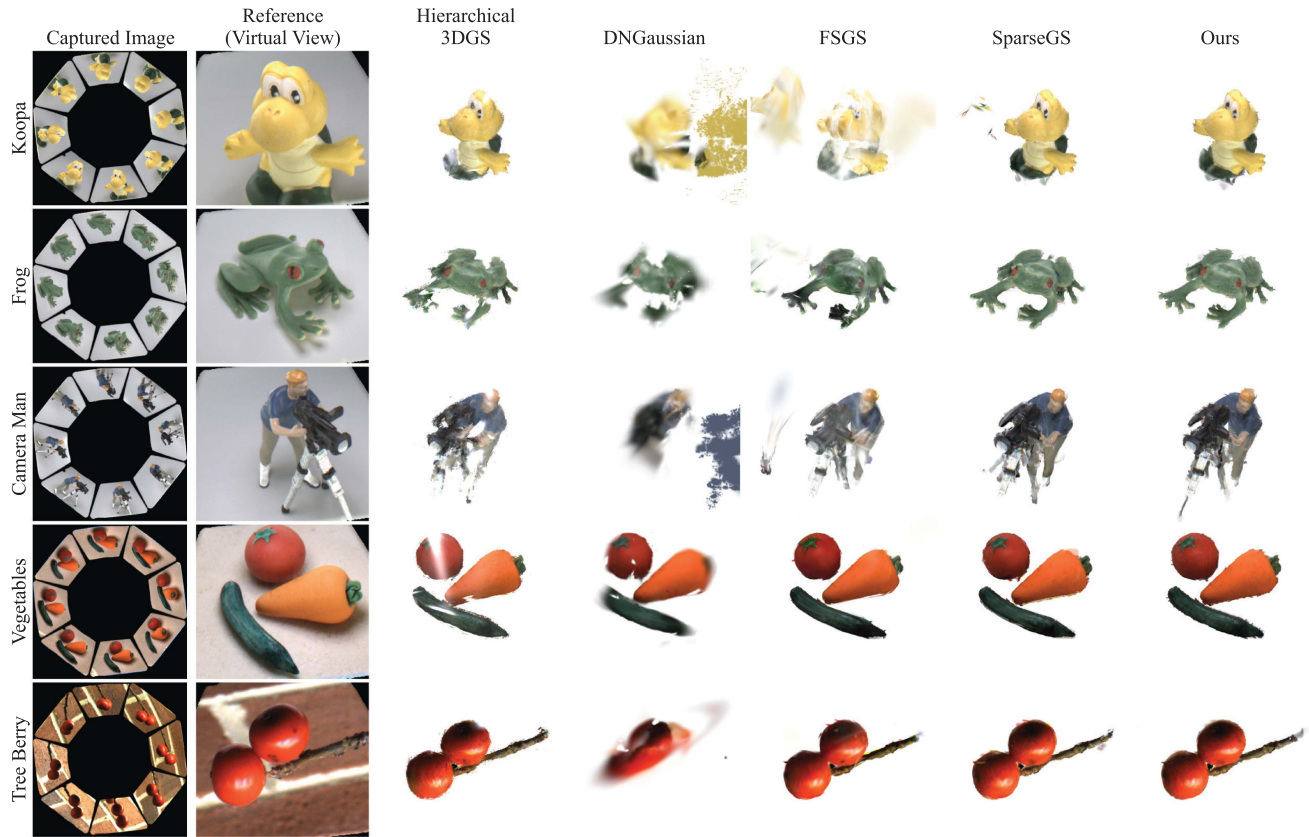


Fig. 15. Visual comparison results on real data. See appendix, available online, for more results and videos of full-surround rendering.

We try lenses with different mirror angles and spacings. Fig. 8 shows comparison between two mirror lenses with different mirror configurations. Specifically, the two lenses have the same base length for the inner and outer pyramids. But the tilting angles for the two mirrors in a pair are different. For prototype a, we have $\alpha_1 = 75^\circ$ (for inner mirror) and $\alpha_2 = 85^\circ$ (for outer mirror), with $\Delta\alpha = 10^\circ$. For prototype b, we have $\alpha_1 = 60^\circ$ and $\alpha_2 = 85^\circ$, with $\Delta\alpha = 25^\circ$. Prototype b's mirror pair has greater tilting angle difference. Images taken with the two lenses are shown in Fig. 8 with one reference view zoomed in.

We can see that prototype b, which has greater tilting angle difference (i.e., $\Delta\alpha$), has better coverage of the side views (e.g., the figurine's face becomes visible in prototype b's image). This is equivalent to having virtual cameras with more oblique angles. Such configuration is preferred for full-surround imaging since it provides fuller coverage of the scene. This observation is consistent with our derived guidelines on optimizing the mirror configuration, shown in Section IV-B.

Since acrylic mirror has limited optical quality that may result in blurriness and aberration in reflection images, we order a customized mirror lens made with high-precision optics from a lens maker (see Fig. 7(d)). The mirror parameters are similar to the prototype b. The lens is more compact in size and thus portable. Its reflection image is sharp and has high quality. We use this lens for capturing real scene data. We use the same 5 megapixel FLIR camera with 12 mm lens for image acquisition. Since the lens is compact and portable, we capture miniature scenes from both indoor and outdoor. We also capture

two dynamic scenes and demonstrate freeze-time novel view synthesis. The sizes of miniature objects in our scene are in the range of 0.5 cm to 5 cm.

B. Implementation Details

Image pre-processing. Fig. 9 shows examples of our captured raw image and processed multi-view images. The raw image captured by our portable lens prototype is shown in Fig 9(a). Its resolution is 2448×2048 . We first apply a multi-view mask to extract the effective regions formed through mirror reflection. The filtered image is shown in Fig. 9(b). Then, for each sub-image, we re-project it to allow smooth view transition: we enforce all virtual cameras to look at the center of the scene and all virtual images have the same up direction. We update camera poses after re-projection. We also mask out the background and only reconstruct the foreground objects. Examples of processed sub-view images are shown in Fig. 9(c)–(f). Each sub-view image has resolution 800×800 , and they are used for initial point cloud estimation and as input to 3DGS.

Pre-calibration. We pre-calibrate our imaging system to obtain accurate intrinsic and extrinsic parameters for the virtual cameras.

Although the calibration can be done with planar calibration target, due to limited viewing volume of our system, it is hard to vary the pose of calibration board. We then made a special calibration target as shown in Fig. 10 for more convenient calibration: we only need to move target by rotation or lifting

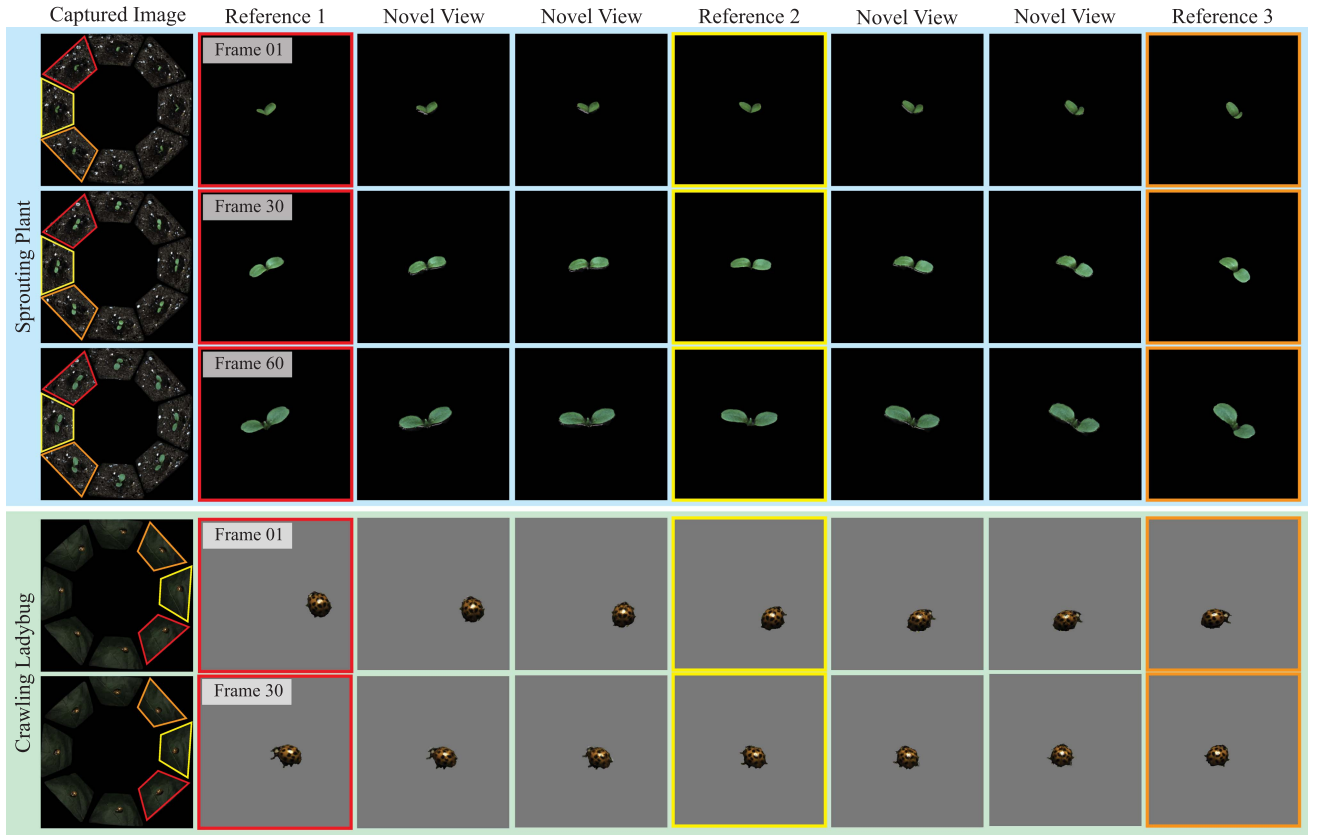


Fig. 16. Novel view synthesis results on dynamic scenes. For each scene, we show captured images, three re-projected reference views (border colors indicate correspondences in the all-in-one image), and novel views synthesized in between. We also show results for multiple time frames.

TABLE III

SENSITIVITY ON CALIBRATION ERRORS. WE APPLY DIFFERENT LEVELS OF ANGULAR PERTURBATION TO ALL VIRTUAL CAMERAS' ORIENTATIONS.

Maximum angular perturbation	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
0° (no error)	0.9713	30.7049	0.0262
0.05°	0.9052	24.5984	0.0548
0.1°	0.8879	22.9681	0.0770
0.25°	0.8549	19.4486	0.1234

up/down. We use contrast colors (e.g., black and white) to distinguish different viewpoints. We use asymmetric circle grids to eliminate the 180-degree rotation ambiguity. Fig. 10(b) shows a captured image of the target with detected circle centers.

Then we run standard calibration algorithm for asymmetric circle grid [44] to estimate intrinsic and extrinsic parameters for the virtual cameras. Since mirror reflection doesn't change intrinsic parameters, we first estimate the intrinsics with all available views and use the same set of parameters for all virtual cameras. We then calibrate the extrinsic parameters after mounting the mirror lens. We also run bundle adjustment to further refine the camera poses.

Since our portable mirror lens is attached to the camera, this calibration process only needs to run one time. The calibration accuracy is affected by the circle detection accuracy and the number of target poses. In our experiment, we use around 30 poses by adjusting the location and orientation of the calibration

target. We exclude images whose circle detection rate or accuracy is too low. Once calibrated, the camera can be carried around for imaging, since the mirror lens is tightly fastened with screw mount. When changing refractive lens or camera, the mirror lens needs to be detached and the system needs to be re-calibrated.

Code implementation. We implement the pre-processing steps, including image re-projection, visual hull generation and visual hull depth rendering *etc.*, using MATLAB and Python. With our multi-view images and calibrated camera parameters, we use Agisoft Metashape⁴ to generate the initial point cloud. We then convert the point cloud and camera parameters to COLMAP format, so they can be used by 3DGS. We use the latest Hierarchical 3DGS [43] as our backbone code. We integrate our visual hull-based depth regularization and optimize the rendering using foreground object mask, such that only the foreground is used for optimization. All our codes are run on a computer with NVIDIA 4090 GPU. The average running for reconstructing a scene using 8 reference views (800×800 each) is around 2 minutes.

C. Synthetic Results

We validate our approach on our simulated data, and compare our results with recent state-of-the-art 3DGS algorithms: Hierarchical 3DGS [43], FSGS [45], DNGaussian [46] and SparseGS [47]. Most of these methods are optimized

⁴<https://github.com/agisoft-llc>

for taking sparse view input and rely on depth prior for optimization. Specifically, Hierarchical 3DGS [43] upgrades the original 3DGS [3] with a divide-and-conquer approach to improve training efficiency. It also integrates the depth regularization. We use this method as our backbone framework. FSGS [45] uses proximity-guided Gaussian unpooling combined with monocular depth map to place new Gaussian between existing ones. DNGaussian [46] adopts a soft depth regularization that freezes Gaussian centers for opacity optimization. SparseGS [47] uses a similar framework as FSGS, but introduces an “unseen viewpoint” regularization to mitigate overfitting.

It is worth noting that since our scenes are small and often lack distinct features, COLMAP fails to run on most of our scenes. For fair comparison, we use our pre-calibrated camera parameters and initial point cloud as input for all comparison methods. For each scene, we use 8 reference views as input and render 24 novel views. For Hierarchical 3DGS, we use depth map generated by Depth Anything V2 [48] for depth loss. For other methods, we use the mono-depth model embedded in their own code. Visual comparison results are shown in Fig. 11. Quantitative comparison results using standard metrics (i.e., SSIM, PSNR and LPIPS) are shown in Table I. The metric values are averaged over all rendered novel views for all scenes. We can see that our method outperforms the state-of-the-arts in both qualitative and quantitative comparisons. The other sparse view methods do not perform very well, likely because they lack the more effective visual hull depth regularization.

Ablation on depth loss. Fig. 12 compares depth maps obtained by different methods for the “frog” scene. Specifically, the MiDaS [49] depth is used by FSGS [45]; Depth Anything V2 [48] is used by Hierarchical 3DGS [43]; and our method uses visual hull depth. We can see that Depth Anything provides much better depth prior than MiDaS depth. Our visual hull depth outperforms Depth Anything result in details (e.g., the frog legs have more discernible depth variation in the visual hull depth). Moreover, the visual hull projection provides depth values in absolute scale, whereas the other two learning-based methods estimate relative depths.

We then performed an ablation study on depth loss using images of three synthetic scenes (“skull”, “squirrel”, and “flower”; see Fig. 11 for reference). We compare three variants of our algorithm: without depth loss (“w/o depth”), with monocular depth obtained by Depth Anything V2 [48] (“w. mono”), and with visual hull (VH) depth as proposed in our approach (“w. VH”). In addition to our own variants, we also test SparseGS [47] using monocular depth versus our visual hull depth on the same set of scenes, to evaluate the generality of the visual hull depth regularization. Table II shows the PSNR of novel view synthesis results for variants of SparseGS and our method using different depth regularization strategies. We can see that depth loss helps improve PSNR in general. Visual hull depth further improves the reconstruction quality over monocular depth when the scene has few texture (e.g., “skull”). But, for objects that are largely concave (e.g., “flower”), monocular depth outperforms visual hull depth, since it better restores the details in the concave region. The visual hull depth can be used as an effective depth

regularization for 3D Gaussian optimization in general when silhouette masks are available.

Number of virtual views. Different numbers of mirror pairs (or virtual views) results in trade-off between the spatial and angular sampling of the catadioptric imaging system. Fig. 13 illustrates the trade-off by showing example images with different numbers of virtual views.

We can see that when using relatively fewer number of mirror pairs (e.g., 4 views), each mirror can have relatively larger size, which results in each virtual camera to have a wider FoV (i.e., large angular coverage). However, the spatial sampling is rather sparse, resulting limited overlap between neighboring views. This poses challenge on the initial multi-view-based point cloud reconstruction, as well as the visual hull carving.

Increasing the number of mirror pairs makes the spatial sampling denser, but the FoV of each virtual camera reduces as the mirror size needs to be smaller (assuming the same overall form factor). A FoV that is too small would result in incomplete scene coverage and weak per-view constraint. From a practical point of view, narrow FoV also poses challenge on calibration.

In our setting of miniature scene reconstruction, we choose to use 8 views as a balanced configuration. But the optimal number of views may vary for scenes of different sizes. For example, if a bigger catadioptric lens is to be constructed for handling larger scenes, more number of mirror pairs (or virtual views) may be desired.

Sensitivity to calibration errors. Here we perform an experiment that evaluates our system’s sensitivity to calibration errors. Specifically, we add random angular perturbations to the virtual cameras’ ground-truth orientations to generate extrinsic parameters with errors. We test on perturbation angles in the range of 0.05° , 0.1° , and 0.25° . We perform the experiments on three synthetic scenes (“Figurine”, “Flower”, and “Squirrel”). The averaged quantitative evaluation metrics are shown in Table III. We can see that the reconstruction quality goes down as the angular perturbation increases. Therefore, accurate camera parameters are critical to high-quality reconstruction.

D. Real Results

We perform comparisons on real scene images taken with our mirror lens. Fig. 15 shows visual comparison results against the state-of-the-arts. We also show our captured image (processed with the multi-view mask) and one re-projected virtual view, which is used as input reference image for 3DGS (we use eight reference views in total). All the scenes are composed with miniature objects with size between 1 cm and 5 cm. The “tree berry” scene is captured outdoor with our portable lens. Same as the synthetic experiments, all comparison methods take our pre-calibrated camera parameters and initial point cloud, since COLMAP fails on those miniature scenes. We apply a foreground mask on each virtual view and only recover the miniature object in foreground. The “camera man” scene is challenging with thin structures around 1 mm (e.g., the tripod legs). Our method is able to recover the thin legs, whereas all other methods cannot.

Comparison with COLMAP-free method. Since camera parameters are critical to the quality of novel view synthesis, we also compare our method against a recent COLMAP-free method, InstantSplat [26], which is able to take sparse-view input. The method bypasses COLMAP by jointly optimizing Gaussian attributes and camera parameters. Visual comparison of a synthesized novel view on a real scene data is shown in Fig. 14. We also show their estimated camera poses in comparison with our calibrated ones. It is challenging to estimate accurate pose with sparse full-surround images (due to limited overlaps). Inaccurate camera parameters then result in low quality novel view synthesis.

Dynamic scene results. We capture two dynamic scenes that are very challenging to reconstruct using other methods: a sprouting plant and a crawling lady bug. The size of the plant is between 2 cm to 4 cm. The ladybug is around 0.5 cm in diameter. For the “sprouting plant”, we take time-lapse photos using our system at one frame per ten minute over two days. We then select 60 frames among 288 frames for novel view synthesis, in order to modeling the growth the plant. For the “crawling ladybug”, we record a one minute video at 60 frames per second, and use all 60 frames for novel view synthesis. The ladybug scene is especially challenging, since the bug is moving fast and multiple cameras needs to be used for 3D reconstruction, however it is hard to setup multiple cameras for macro imaging. We show novel view synthesis results for the two dynamic scenes in Fig. 16.

VII. CONCLUSION AND DISCUSSION

We have presented an imaging solution using circularly arranged mirror pairs for full-surround reconstruction and novel view synthesis of miniature scenes. We have first shown how to configure a pair of planar mirrors to obtain reflection image from a desired viewpoint, and then thoroughly analyzed the design factors of our mirror lens and derived optimal parameters given the scene size. We have tailored the latest 3DGS framework to allow accurate and robust scene reconstruction using our sparse view input. Our method has been validated through synthetic and real experiments and demonstrated state-of-the-art performance. Since our approach is a single-shot solution, we have also demonstrated novel view synthesis results on challenging dynamic scenes.

Our full-surround paired-mirror design has a trade-off between the number of views and the angular resolution of each view: by using more pairs of mirrors on a circle, we are able to obtain more virtual viewpoints, however the angular resolution (e.g., field of view) of one point would decrease. This is why we choose to use eight mirror pairs, which balances between spatial viewpoint sampling and per-view angular coverage. It is possible to combine mirror pairs with coded patterns on sensor to allow overlapped virtual views, in order to obtain denser viewpoint sampling. In terms of handling dynamic scenes, one can incorporate temporal constraints to increase the optimization efficiency and improve the rendering quality.

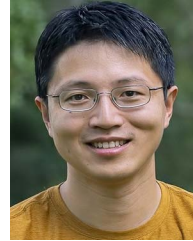
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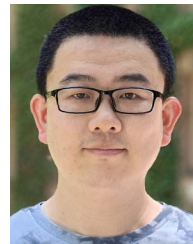
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